

SmallPlaneValue - Tier Pricing & Monetization Spec

Document purpose: Product and engineering reference for implementing paid tiers on smallplanevalue.com.

Audience: Developer implementing billing, rate limits, and feature gating.

Date: June 2026

Status: Proposal - not yet implemented in code.

Executive summary

SmallPlaneValue should use a freemium + credits model, not a subscription-only model. Most users need valuations once per aircraft hunt; recurring subscriptions fit only active buyers and brokers.

Recommended launch: Free tier + \$24 single full valuation + \$79 five-pack. Add subscriptions after repeat usage is proven.

Positioning: *"\$25 to sanity-check a \$300k decision."* Paid users get more depth and tools, not inflated numbers. Trust is the product.

Current product surface (what exists today)

Feature	Route / location	API cost profile
N-number lookup	<code>`/api/faa-lookup`</code> , Lookup tab	Low (Supabase)
AI valuation	<code>`/api/valuate`</code> , Valuation tab	High (Anthropic + web search + comps)
Market comps	<code>`/api/comps`</code> , Comps tab	Medium-High
Pre-buy checklist	<code>`/api/checklist`</code> , Checklist tab	Medium
Listing auto-fill	<code>`/api/parse-listing`</code>	Medium
Sold-price submit	Sold tab (localStorage today)	Low
Feedback	<code>`/api/feedback`</code>	Low

Funnel: ntailnum.com ? smallplanevalue.com (shared Supabase aircraft DB).

Pricing tiers

Tier 1 - Free (discovery + funnel)

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|---|---|
Price | \$0
Goal | SEO, ntailnum funnel, trust-building

Includes

- N-number lookup - unlimited
- Basic valuation - **1-2 per month** per IP or account
- Simplified output - seller ask + fair market only (hide negotiating tips, comp detail, confidence breakdown)
- Pre-buy checklist - **1 per month**
- Sold-price submission - unlimited (feeds crowd-sourced transaction database)

Excludes / limits

- Full valuation report (buyer target, negotiating tips, detailed analysis)
- PDF export
- Saved aircraft history
- Unlimited comps

Tier 2 - Pilot Pass (pay-per-report)

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|---|---|
Price | **\$19-29 per full valuation** (recommend **\$24** at launch)
Bundle | **5-pack for \$79**
Goal | Casual buyers shopping one aircraft

Includes

- Full valuation: seller ask, fair market value, buyer target
- Avionics / logbook / damage / out-of-annual adjustments (existing deterministic post-processing)
- Negotiating tips + confidence score
- Market comps unlocked for that aircraft
- Pre-buy checklist for that aircraft
- PDF export or shareable link (to be built)

Stripe product suggestion

- `spv\_valuation\_single` - \$24 one-time

- `spv\_valuation\_5pack` - \$79 one-time (5 credits)

Tier 3 - Buyer Pro (subscription)

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Price | **\$39/month** or **\$299/year**

Goal | Active shoppers comparing 3-10 aircraft

Includes

- **10 full valuations per month**
- Unlimited comps + checklists
- Listing paste auto-fill
- Save / compare aircraft side-by-side (to be built)
- Email alerts when sold-price data is added for watched models (future)

Stripe product suggestion

- `spv\_buyer\_pro\_monthly` - \$39/mo
- `spv\_buyer\_pro\_annual` - \$299/yr

Tier 4 - Broker / CFI / Shop (B2B)

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|---|---|

Price | **\$99-149/month** or **\$999/year**

Goal | Users running valuations weekly

Includes

- **30-50 valuations per month**
- White-label PDF ("Prepared with SmallPlaneValue")
- Client-facing share links
- Priority support
- API access (future, optional)

Stripe product suggestion

- `spv\_broker\_monthly` - \$129/mo (midpoint)
- `spv\_broker\_annual` - \$999/yr

Optional add-ons (future)

Add-on | Price | Description

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Listing review | \$9-15 | Paste a Controller / TAP listing; get over/under-ask verdict

Negotiation brief | \$9-15 | One-page email template for seller negotiation

Feature gating matrix

Feature | Free | Pilot Pass (credit) | Buyer Pro | Broker

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N-number lookup | [x] unlimited | [x] | [x] | [x]

Basic valuation (limited fields) | [x] 1-2/mo | - | - | -

Full valuation | - | [x] per credit | [x] 10/mo | [x] 30-50/mo

Comps (full detail) | Limited | [x] per aircraft | [x] unlimited | [x] unlimited

Checklist | 1/mo | [x] per aircraft | [x] unlimited | [x] unlimited

Parse listing | - | [x] | [x] | [x]

PDF / share link | - | [x] | [x] | [x] white-label

Sold-price submit | [x] | [x] | [x] | [x]

Saved / compare aircraft | - | - | [x] | [x]

What to gate in code (engineering notes)

High-cost endpoints - require auth + credit or active subscription

- `POST /api/valuate` - full response unless free-tier basic mode
- `POST /api/comps` - full detail behind paywall
- `POST /api/checklist` - beyond free monthly limit
- `POST /api/parse-listing` - Buyer Pro+ or consumes a credit

Keep free (low cost, high funnel value)

- `POST /api/faa-lookup`
- `POST /api/feedback`

- Sold-price submission UI (move from localStorage to Supabase when ready)

Free-tier valuation behavior

When a free user hits /api/valuate, either:

1. Return truncated JSON (sellerAsk + fairMarketValue only), or
2. Return full JSON but hide fields in UI with upgrade CTA

Recommend server-side truncation so API cannot be scraped for full reports.

Rate limiting

- Free: 1-2 valuations / month per `user\_id` or fingerprinted IP
- Track usage in Supabase table: `usage\_events` (user_id, feature, created_at)
- Credits: `user\_credits` (user_id, balance, updated_at) decrement on full valuation

Auth

- Email magic link or Google OAuth (Supabase Auth fits existing stack)
- ntailnum.com can share auth session or deep-link with token later

Payments

- **Stripe Checkout** for one-time packs and subscriptions
- Webhook: `checkout.session.completed` ? add credits or activate subscription
- Store: `subscriptions` (user_id, stripe_customer_id, plan, status, period_end)

Launch sequence (phased rollout)

Phase 1 - Now

- Free lookup + **1 free full valuation per month** (email capture before 2nd valuation)
- No Stripe yet; measure conversion intent (click "Get full report")

Phase 2 - Week 2

- Stripe: **\$24 single report** + **\$79 five-pack**
- Gate `/api/valuate` full response behind credit check
- Add upgrade CTA on valuation results page

Phase 3 - Month 2

- Buyer Pro subscription (\$39/mo, \$299/yr)
- PDF export + saved aircraft list
- Only launch if analytics show repeat users

Phase 4 - Month 3+

- Broker tier
- White-label PDF
- Consider API for partners

Do not launch all four tiers on day one. Start with Free + \$24/report.

Competitive positioning

Competitor | Their angle | SmallPlaneValue counter

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Sandhill (if/when live) | Likely black-box number | Show your work - comps, adjustments, buyer vs seller view

Vref / Aircraft Bluebook | Dealer reference subscription | Pilot-friendly, one-off pricing, experimental/kit coverage

ASA certified appraisal | Legal / financing grade | Pre-transaction research - "not an appraisal" is a feature

Disclaimer (already on site): Not a certified appraisal. For financing, insurance, or legal transactions, engage an ASA-certified appraiser.

Pricing rationale (for stakeholder context)

- Pre-buy inspection: **\$500-\$1,500**
- Certified appraisal: **\$500-\$2,000+**
- Vref dealer access: **~\$100-300/year**
- SPV at **\$24/report** is an easy yes if it prevents a \$20k overpay on a \$300k aircraft
- Subscription only makes sense for users actively comparing multiple aircraft

Trust rule (non-negotiable)

Never charge for inflated valuations. Paid tiers unlock depth (comps, tips, PDF, history), not a higher

number. This is core brand differentiation vs black-box competitors.

Open questions for dev

1. Supabase Auth vs separate auth provider?
2. Credit expiration (e.g. 12 months) for 5-packs?
3. Guest checkout (email only) vs account required?
4. Move sold-price data from localStorage to Supabase now or with billing?
5. PDF generation: server-side (Puppeteer) vs client print stylesheet?

Suggested Supabase tables (starter schema)

```
-- users extended via Supabase Auth (auth.users)

create table user_credits (
  user_id uuid primary key references auth.users(id),
  balance int not null default 0,
  updated_at timestampz default now()
);

create table subscriptions (
  id uuid primary key default gen_random_uuid(),
  user_id uuid references auth.users(id),
  stripe_customer_id text,
  stripe_subscription_id text,
  plan text not null, -- 'buyer_pro' | 'broker'
  status text not null, -- 'active' | 'canceled' | 'past_due'
  period_end timestampz,
  created_at timestampz default now()
);

create table usage_events (
  id uuid primary key default gen_random_uuid(),
  user_id uuid references auth.users(id),
  feature text not null, -- 'valueate' | 'comps' | 'checklist' | 'parse_listing'
  metadata jsonb,
  created_at timestampz default now()
);

create index usage_events_user_feature_created
on usage_events (user_id, feature, created_at desc);
```

Contact / owner

Product owner: John Wallace

Site: <https://smallplanevalue.com>

Related: <https://ntailnum.com> (N-number lookup funnel)

End of document.